GEOMETRY BY CONSTRUCTION: A GEOMETRIC PRIOR BOTTLENECK

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ABSTRACT

The information bottleneck (IB) principle seeks representations that retain target information while discarding input variation. The conditional entropy bottleneck (CEB) uses a label-conditional reference, but its class-wise matching terms leave the geometry between different labels unspecified. We introduce the geometric prior bottleneck (GPB) framework, which declares this geometry: OPB uses an orthogonal class frame for classification and EPB an isometric label axis for regression. In both cases, the same geometric means are used as conditional-Gaussian KL targets and prediction anchors through a parameter-free energy or projection head. GPB preserves the CEB certificate and yields comparator-dependent alignment and separation guarantees whose combination gives an explicit β-dependent geometry-transfer bound with a comparator-specific floor. Controlled label-noise, direct mismatch, robustness, and posterior-geometry studies characterize these mechanisms, while the full benchmark comparison spans twelve dataset–backbone settings and complete IB and non-IB baselines. GPB is best or tied-best in seven settings and has a higher mean than CEB in all twelve.

1 INTRODUCTION

A useful representation must balance compression and structure: it should discard input variation irrelevant to the target while preserving target-relevant information in a geometry that supports reliable prediction.

Geometric approaches focus on how target information is organized within a representation. Across classification and regression, neural-collapse-inspired frames, equidistant or orthogonal constructions, fixed classifiers, and isometric embeddings or readouts impose structure on features, prototypes, model weights, or prediction axes (Papyan et al., 2020; Deng et al., 2014; Xu et al., 2022; Tong & Davis, 2023; Koromilas et al., 2026; Pai et al., 2022; Luo et al., 2023; Belucci et al., 2026). These methods make target geometry explicit, but their objectives primarily shape the structure needed for prediction. They do not generally include a separate criterion for controlling the target-irrelevant input variation retained in the representation, leaving compression implicit.

Information-bottleneck methods such as VIB, NIB, and DVCCA make compression explicit by limiting the information a representation retains from the input through marginal-prior matching, prior-free estimation, or reconstruction-augmented objectives (Alemi et al., 2017; Kolchinsky et al., 2019; Abdelaleem et al., 2025). Despite their different formulations, their compression mechanisms are not conditioned on the target label and therefore do not isolate the input variation retained beyond it. CEB addresses this limitation by introducing a label-conditional reference and directly targeting the residual input information that remains after the label is known (Fischer, 2020).

Conditioning the reference on the label does not by itself determine the relations between different labels. The CEB certificate is a sum of within-label matching terms, with no term coupling two distinct label conditionals: it records neither the relative directions nor the pairwise distances of their means. Consequently, the inter-label arrangement is determined by optimization dynamics and the freedom of the prediction head rather than by the compression criterion. This motivates making cross-label geometry part of the conditional prior itself.

We introduce the geometric prior bottleneck (GPB) framework, which places a declared geometry directly in the label-conditional reference. A GPB prior is a conditional Gaussian with bounded covariance whose means satisfy a prescribed separation structure for discrete labels or a prescribed label metric for continuous targets. The form and scale of this geometry are chosen before training, while its overall orientation can remain learnable. We instantiate GPB in two forms. For classification, the orthogonal prior bottleneck (OPB) constructs an equal-radius orthogonal frame and predicts using the anchor energies induced by that frame. For regression, the equidistant prior bottleneck (EPB) maps labels onto an isometric axis and predicts through a projection tied to the same axis. In both cases, the same prior means serve as KL targets and prediction anchors, unifying information compression with explicit target geometry.

Our analysis connects the declared prior geometry to the learned posterior. Theorem 2.1 bounds the average anchor mismatch between posterior means and their prescribed anchors by a comparator-dependent certificate. Theorem 2.2 turns measured mismatch into a lower bound on the distance between posterior centers of different classes. Corollary 2.3 combines the two into the main end-to-end guarantee: with zero comparator KL floor, the certified deficit from the declared separation is $O(\beta^{-1/2})$, while a non-zero comparator KL imposes a comparator-specific floor.

We evaluate GPB across twelve dataset–backbone settings spanning classification and regression. GPB is best or tied-best in seven settings and achieves a higher mean than CEB in all twelve. Controlled studies examine the comparator-dependent floor, while geometric diagnostics trace β-dependent mismatch and transfer of the declared structure to posterior class centers. Compression sweeps and targeted robustness experiments reveal favorable behavior in regimes where CEB degrades, and factor-wise ablations distinguish the contribution of the geometric bottleneck from that of the readout.

In summary, our contributions are: (i) identifying and formalizing the missing cross-label geometric constraint in CEB; (ii) introducing the GPB framework, with OPB and EPB, to unify conditional compression with explicit target geometry; (iii) establishing an upper bound on average anchor mismatch and a lower bound on posterior class-center separation, together with an explicit β-dependent geometry-transfer rate; and (iv) demonstrating the method across classification and regression through benchmark comparisons, geometric diagnostics, compression studies, robustness experiments, and factor-wise ablations.

2 METHOD

2.1 CONSTRUCTING THE GEOMETRIC PRIOR FAMILY

Let $Y - X - Z$ and let the encoder posterior be $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \operatorname{diag}\sigma^2_\phi(x))$. CEB (Fischer, 2020) replaces the VIB marginal reference by a label-conditional reference $r(z|y)$. For any choice of this reference, the certificate

$$I(X;Z|Y) \le \mathbb{E}\,\mathrm{KL}(q_\phi(z|X)\|r(z|Y)), \tag{1}$$

upper-bounds the residual input information $I(X;Z|Y)$, which CEB minimizes as its compression criterion. The certificate is an average of within-sample terms, each involving only the reference of the sample's own label: no term couples two distinct label conditionals. Consequently, it records neither the relative directions nor the pairwise distances of the class-conditional means, so the compression criterion alone cannot prefer any cross-label geometry.

The GPB framework. GPB is not a single prior but a framework for declaring reference geometry: it restricts the reference to conditional Gaussians

$$p_\theta(z|y) = \mathcal{N}(m_\theta(y), \Sigma_\theta(y)), \qquad \Sigma_\theta(y) \preceq \tau^2_{\max} I, \tag{2}$$

where p_θ is the label-conditional Gaussian reference, $m_\theta(y)$ its anchor mean, $\Sigma_\theta(y)$ its anchor covariance, and $\tau^2_{\max}$ a fixed bound on the anchor scale. Their means obey a declared two-sided separation for every pair of distinct discrete labels $y \neq y'$,

$$\Delta \le \|m_\theta(y) - m_\theta(y')\| \le \bar{\Delta}, \tag{3}$$

where Δ and $\bar{\Delta}$ are the declared minimal and maximal anchor distances (the separation margin and its ceiling), or a two-sided label metric condition for P_Y-almost every label pair (y, y'), where d_Y is a metric on the label space $\mathcal{Y}$:

$$\rho_{\min} d_Y(y,y') \le \|m_\theta(y) - m_\theta(y')\| \le \rho_{\max} d_Y(y,y'), \tag{4}$$

so the anchor map $y \mapsto m_\theta(y)$ is a bi-Lipschitz embedding of the label metric, with lower and upper constants $\rho_{\min}$ and $\rho_{\max}$. All constants are fixed before training. These conditions define the geometric prior family $\mathcal{G}$. The objective is

$$\mathcal{L}_{\mathrm{GPB}} = \mathbb{E}[\ell(c_\psi(z), Y) + \beta\,\mathrm{KL}(q_\phi(z|X)\|p_\theta(z|Y))], \qquad z \sim q_\phi(z|X). \tag{5}$$

The instances studied below, OPB (classification) and EPB (regression), are two members of this framework, both realizing the single objective above: the anchor means enter the KL and are reused as prediction anchors through the tied readout (anchor-energy scores for classification, tied projection for regression), so the declared geometry is shared by compression and prediction and no instance-specific objective is stated. Here $\ell \geq 0$ is cross-entropy for classification and the chosen nonnegative regression loss for continuous labels. Moreover, since the geometric prior remains a label-conditional reference, the CEB certificate of Eq. (1) carries over unchanged, unifying compression with the declared geometry in a single KL term, and any method whose reference prior belongs to $\mathcal{G}$ inherits the guarantees of Section 2.2.

Instances. For classification, OPB evaluates a prior encoder on all class labels and uses its thin polar factor Q_θ. If U_θ is the resulting all-class matrix, then

$$Q_\theta = U_\theta(U_\theta^\top U_\theta)^{-1/2}, \qquad Q_\theta^\top Q_\theta = I_K, \tag{6}$$

and we assume, monitoring rather than imposing it, that the monitored matrix has full column rank (Assumption A.1 in the appendix). Writing $q_{\theta,k}$ for the k-th column of the frame Q_θ, the anchors and their fixed geometry are

$$p_\theta^{\mathrm{OPB}}(z|Y=k) = \mathcal{N}(aq_{\theta,k}, \tau^2 I), \quad (aq_i)^\top(aq_j) = a^2\mathbf{1}\{i=j\}, \quad \|aq_i - aq_j\| = \sqrt{2}a\ (i \neq j). \tag{7}$$

The anchor-energy scores are the Gaussian Bayes scores

$$s_k(z) = -\frac{\|z - aq_{\theta,k}\|^2}{2\tau^2} + \log \pi_k, \tag{8}$$

With uniform priors, this induces an orthogonal classifier whose scale is fixed by the prior (Appendix A.3). Detailed polar construction, initialization, covariance relaxation, and the relation to prototype/ETF methods are given in Appendices A.3 and C.

For regression, EPB uses a one-dimensional continuous label, standardized once on the training set as $\tilde{y}$ with the statistics frozen. Let $W \in \mathbb{R}^{d\times 1}$ be a trainable direction with per-step normalization $u = W/\|W\|$; the isometric prior is

$$p_\theta^{\mathrm{EPB}}(z|y) = \mathcal{N}(\rho\tilde{y}\,u, \tau^2 I), \qquad \|\mu_p(y) - \mu_p(y')\| = \rho|\tilde{y} - \tilde{y}'|, \tag{9}$$

so latent distances equal ρ times standardized label distances exactly. Order and magnitude of label differences are encoded with equality, which makes it a member of $\mathcal{G}$ (Eq. (4) with $\rho_{\min} = \rho_{\max} = \rho$). The prediction head is the tied projection $\hat{\tilde{y}} = u^\top z/\rho$, the regression analogue of the anchor-energy classifier: no free direction or scale can bypass the prior.

2.2 Comparator bounds and posterior-center separation

This subsection traces how the objective of Eq. (5) shapes the learned posterior: Theorem 2.1 upper-bounds the posterior's anchor mismatch, Theorem 2.2 lower-bounds the separation of posterior centers and introduces the condition under which the bound is informative, and Corollary 2.3 combines the two into the end-to-end β-dependent transfer guarantee. Throughout, $\Delta(y, y')$ denotes the declared lower bound on the anchor pair distance: $\Delta(y, y') = \Delta$ for distinct discrete labels (Eq. (3)) and $\Delta(y, y') = \rho_{\min} d_Y(y, y')$ for continuous labels (Eq. (4)). Define

$$\mathcal{R}(w) = \mathbb{E}[\ell(c_\psi(Z), Y)], \qquad \mathcal{K}(w) = \mathbb{E}\,\mathrm{KL}(q_\phi(z|X)\|p_\theta(z|Y)), \tag{10}$$

and

$$\mathcal{D}(w) = \frac{\mathbb{E}\|\mu_\phi(X) - m_\theta(Y)\|^2}{2\tau_{\max}^2}. \tag{11}$$

Theorem 2.1 (Comparator-dependent bound on anchor mismatch)**.** *Let $\beta > 0$, $\varepsilon \geq 0$, and $\ell \geq 0$. If $\widehat{w}$ is ε-optimal for $\mathcal{R} + \beta\mathcal{K}$, then for every feasible comparator w_0 in the same family (same encoder, posterior, prior, and prediction-head parameterization),*

$$\mathcal{D}(\widehat{w}) \leq \mathcal{K}(\widehat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta}. \tag{12}$$

The bound certifies alignment to whatever comparator is prespecified: its KL $\mathcal{K}(w_0)$ is the comparator-specific certificate floor, a property of the chosen comparator rather than an intrinsic dataset quantity, and the risk term controls how fast the right-hand certificate bound approaches that floor as β grows. Equivalently, to certify a target mismatch level $d_\star$, one needs a comparator with $\mathcal{K}(w_0) < d_\star$ and $\beta \geq (\mathcal{R}(w_0) + \varepsilon)/(d_\star - \mathcal{K}(w_0))$.

Theorem 2.2 (Mismatch-conditioned posterior-center separation)**.** *Suppose $\mathbb{E}\|\mu_\phi(X)\|^2 < \infty$ and, for a label value y, let*

$$m(y) = \mathbb{E}[\mu_\phi(X) \mid Y = y], \qquad D(y) = \frac{\mathbb{E}[\|\mu_\phi(X) - m_\theta(y)\|^2 \mid Y = y]}{2\tau_{\max}^2}. \tag{13}$$

Then, for every pair of distinct discrete labels and for $P_Y \otimes P_Y$-almost every continuous pair (y, y'),

$$\begin{aligned} \|m(y) - m(y')\| &\geq \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')} \\ &\geq \Delta(y, y') - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')}. \end{aligned} \tag{14}$$

The first-level bound is positive exactly when

$$\Gamma_{yy'} := \frac{\sqrt{2\tau_{\max}^2 D(y)} + \sqrt{2\tau_{\max}^2 D(y')}}{\|m_\theta(y) - m_\theta(y')\|} < 1, \tag{15}$$

while the declared-margin level is positive exactly when the measured correction terms stay below the declared bound, $\sqrt{2\tau_{\max}^2 D(y)} + \sqrt{2\tau_{\max}^2 D(y')} < \Delta(y, y')$, which in the continuous case fails most readily for nearby label pairs. For OPB the two conditions coincide, since the anchor pair distance is the constant $\sqrt{2}a$ of Eq. (7): $\Delta(y, y') = \sqrt{2}a$ and, with equal mismatch D, positivity holds exactly for $D < a^2/(4\tau_{\max}^2)$.

The anchor scale is a validation-selected design parameter and does not itself guarantee a positive bound: the guarantee is informative exactly when the measured mismatch stays below the declared margin, and small measured mismatch transfers the declared geometry to the posterior centers almost intact.

Corollary 2.3 (Beta-dependent transfer of anchor separation)**.** *Fix the prior family, a feasible comparator w_0, and an ε_β-optimal solution $\widehat{w}_\beta$ for each $\beta > 0$, with $\sup_\beta \varepsilon_\beta < \infty$. Define*

$$B_\beta := \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon_\beta}{\beta}. \tag{16}$$

For any label set S and level $\bar{D}_\beta$ with $D(y) \leq \bar{D}_\beta$ for every $y \in S$,

$$\|m(y) - m(y')\| \geq \Delta(y, y') - 2\sqrt{2\tau_{\max}^2 \bar{D}_\beta} \qquad \textit{for distinct } (y, y') \in S^2. \tag{17}$$

For discrete labels, suppose $\pi_k \geq \pi_{\min} > 0$ for all considered classes; then $\bar{D}_\beta = B_\beta/\pi_{\min}$ covers every class (Theorem 2.1 with the tower property), so Eq. (17) holds for every pair of distinct classes. For continuous labels, for any $\xi \in (0, 1)$, $\bar{D}_\beta = B_\beta/\xi$ covers a set $S_{\beta,\xi}$ of label mass at least $1 - \xi$ (Markov's inequality), so Eq. (17) holds for $P_Y \otimes P_Y$-almost every pair in $S_{\beta,\xi}^2$.

With a fixed comparator satisfying $\mathcal{K}(w_0) = 0$ and the remaining quantities bounded while β varies, the certified separation deficit is $O(\beta^{-1/2})$; with $\mathcal{K}(w_0) > 0$, the certificate approaches the comparator-dependent non-zero upper bound $\mathcal{K}(w_0)$. These rate statements concern the certificate only: they do not claim that the realized mismatch $\mathcal{D}(\widehat{w}_\beta)$ or the realized center distances converge at this rate.

3 Experiments

The evaluation addresses five research questions, each answered by the sections noted below.

(RQ1) End-to-end utility. Does GPB improve the operating point across dataset–backbone settings, and which design factor carries the gain? The main predictive comparison (Sec. 3.2) reports the former, and the paired factor-wise attribution ladder (Sec. 3.6) isolates the latter.

Table 1: Test accuracy (%) on classification tasks and test R^2 on regression tasks (mean over runs); the best entry per dataset is bolded. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines; dashes mark the settings where a method does not apply.

Dataset	Backbone	Base	VIB	SVIB	NIB	CEB	DVCCA	NCBD	AdaCap	GPB	GPB-L
MNIST	MLP	98.33	98.28	98.15	98.27	98.43	98.24	97.95	–	**98.54**	98.43
MNIST	CNN	99.13	99.28	99.23	99.22	99.30	99.29	99.18	–	**99.32**	99.31
ImageNet-100	MLP	83.12	82.84	83.02	82.35	82.77	82.99	83.64	–	**84.40**	84.01
ImageNet-100	CNN	83.40	82.94	83.11	82.91	82.97	83.21	80.56	–	**86.08**	84.88
Cora	GCN	87.68	86.83	86.46	86.35	86.27	86.90	87.34	–	**88.12**	87.86
IMDb	RNN	84.85	85.50	85.99	86.14	85.55	85.69	86.69	–	86.84	**86.99**
AG News	RNN	**92.94**	92.72	92.86	92.91	92.83	92.78	92.37	–	92.87	92.92
Cal. Housing	MLP	0.7618	0.7901	0.7897	0.7862	0.7894	0.7878	–	0.7758	0.7895	**0.7914**
STS-B	RNN	**0.2899**	0.2719	0.2697	0.2417	0.2668	0.2713	–	0.1613	0.2762	0.2755
ZINC	GCN	0.6348	0.6339	0.6325	0.6178	0.6257	0.6294	–	**0.6770**	0.6344	0.6323
AgeDB	MLP	0.2180	0.3173	0.3385	0.2710	0.3317	0.3180	–	0.3482	**0.3536**	0.3320
AgeDB	CNN	0.3914	0.3342	0.3323	0.3337	0.3882	0.3205	–	0.4684	**0.4700**	0.4388
Mean rank		4.83	5.75	5.50	7.00	5.96	6.00	6.43	4.40	1.83	2.54
Best or tied-best		2	0	0	0	0	0	0	1	7	2

(RQ2) Comparator floor. Does the trained posterior attain the theoretically predicted ambiguity floor and never leave the explicit comparator's reachable set? The controlled label-noise study varies a known floor from zero to 11.2 nats and checks the comparator gate (Sec. 3.3).

(RQ3) Geometry transfer and certification. How much of the declared prior geometry reaches the learned posterior, and does the separation guarantee survive the measured mismatch? The direct mismatch analysis measures the transfer, and the structural diagnostics verify the posterior-side structure (Secs. 3.4 and 3.9).

(RQ4) Compression behavior. What do the methods compress and discard along the β sweep, and how does the anchor scale trade accuracy against compression? The compression curves and the information-plane estimates read both (Secs. 3.7 and 3.8).

(RQ5) Robustness. Does the compressed geometric operating point reduce targeted adversarial success without giving up clean accuracy? The white-box PGD study answers it (Sec. 3.5).

3.1 Experimental setup

We evaluate twelve dataset–backbone settings over nine public datasets: MNIST, ImageNet-100, Cora, IMDb, AG News, California Housing, STS-B, ZINC-12k, and AgeDB, covering MLP, CNN, GCN, and RNN backbones. All methods use the same splits, backbone, stochastic Gaussian posterior, bottleneck dimension $d = 256$, Adam training, early stopping, and validation-based selection. The baselines are Base, VIB, SVIB, NIB, CEB, supervised β-DVCCA, and the external NCBD/AdaCap comparisons; GPB is OPB on classification rows and EPB on regression rows, with GPB-L denoting the identical prior and a free head. Full data preprocessing, grids, seeds, and baseline definitions are in Appendix A.3. Regression compression curves, transfer attacks, pair-level mismatch plots, and full uncertainty tables are reported in Appendices B.3, B.5, A.6, and A.7–A.8; the remaining ablations and estimator panels are collected in Appendix B.

3.2 Main predictive comparison

Table 1 reports the best validation-selected configuration of every baseline and GPB variant; the complete standard deviations, paired confidence intervals, and per-method configuration details are in Appendices A.7 and A.8. The ranking summary is dominated by the two GPB heads: GPB has the lowest mean rank (1.83 versus 2.54 for GPB-L and 4.83 for the best plain baseline) and is best or tied-best in 7/12 settings. The gains are largest where the task has margin to resolve them: on ImageNet-100 (CNN) GPB leads CEB by 3.11 points and on Cora by 1.85, while the saturated settings are flat (MNIST CNN and AG News, where all methods sit within 0.6 points of the best) and the small-data regressions are essentially tied with the best baselines (Housing within 0.002 R^2

r	δ_{noise}	$\widehat{D}(\beta{=}10^{-3})$	$\widehat{D}(\beta{=}10)$	max excess	$\widehat{L} - L_{\text{comp}}$
0	0.000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.000	0.000 ± 0.000
0.1	3.400	3.4142 ± 0.0002	3.4018 ± 0.0000	0.014	-0.005 ± 0.004
0.2	6.400	6.4225 ± 0.0005	6.4067 ± 0.0002	0.023	-0.053 ± 0.008
0.4	11.200	11.2324 ± 0.0016	11.2172 ± 0.0003	0.032	-0.191 ± 0.016

Table 2: Controlled label-noise floor (mean $\pm$ std over five seeds); the last column is the fitted objective minus the explicit comparator objective at $\beta = 10$.

of VIB; on STS-B the plain backbone remains best). The tied head costs nothing anywhere except where it helps: it is within 0.4 points of GPB-L in $11/12$ settings, and on ImageNet-100 (CNN) it is 1.20 points better, the setting where the free head has the most capacity to overfit. Among the external non-IB baselines, AdaCap is competitive on regression (best on ZINC), while NCBD's mean rank (6.43) places it behind the plain backbone on classification. Across modalities, the same construction improves the operating point where improvement is available and costs nothing elsewhere.

3.3 Controlled label ambiguity

This experiment isolates the comparator floor by making the label-ambiguity term of the mismatch bound analytically known, and verifies three claims: the noise-free case realizes the zero floor exactly, the measured mismatch tracks the analytic floor at every noise level, and the fitted objective never leaves the explicit comparator's reachable set. The construction uses $K = 10$, $X = e_C$, and symmetric label noise $r \in \{0, 0.1, 0.2, 0.4\}$ with OPB $(a, \tau) = (6, 1)$. The analytic ambiguity floor is

$$\delta_{\text{noise}}(r) = \frac{a^2}{2\tau^2}\left[1 - (1-r)^2 - \frac{r^2}{K-1}\right], \tag{18}$$

which grows from 0 to 11.2 nats as r runs over the noise grid. The explicit comparator takes posterior mean $h_0(x) = a\sum_k \eta_k(x) q_k$ and covariance $\tau^2 I$, where $\eta_k(x) = P(Y = k \mid X = x)$; its KL equals $\delta_{\text{noise}}(r)$ exactly, and its anchor-energy risk is $C_{\text{comp}}(r)$. The training protocol (population-weighted objective, warm start, Monte Carlo risk estimation) is in Appendix A.5. Whenever the fitted objective satisfies $\widehat{L} \le L_{\text{comp}} = C_{\text{comp}} + \beta\delta_{\text{noise}}$, non-negativity of the fitted risk gives the directly checkable comparison

$$\widehat{D} \;\le\; \widehat{\mathcal{K}} \;\le\; \delta_{\text{noise}} + \frac{C_{\text{comp}}}{\beta}. \tag{19}$$

Thus the known-noise experiment tracks the predicted floor, while the sampled-label sensitivity and comparator-gate checks are deferred to Appendix A.5.

Table 2 reads the transition from the zero to the non-zero floor column by column. The zero-noise row reproduces the exact-realizability limit to the stored precision. For $r > 0$ the measured mismatch tracks the analytic floor at both ends of the β grid: at $\beta = 10$ it matches the floor to within 0.02 nats at every noise level, while at the smallest β it sits above the floor by an excess that grows with the noise level (from 0.014 to 0.032 nats) and declines monotonically with compression, without implying an exact $1/\beta$ law. The last column closes the loop: the fitted objective satisfies $\widehat{L} \le L_{\text{comp}}$ at every reported point, and the margin over the explicit comparator grows with the noise level (from -0.005 to -0.191), so the trained posterior moves farther inside the comparator's reachable set exactly where the floor is largest. Sampled-label, fixed-variance, and trainable-frame sensitivity controls are deferred to Appendix A.5.

3.4 Direct mismatch analysis

This section pursues two checks on the learned posterior: how much of the declared anchor geometry it inherits, and whether the separation guarantee survives under the measured mismatch. For each checkpoint, the empirical variance bound $\widehat{\tau}^2_{\max}$ is the largest conditional-reference variance realized over the class table (classification) or test labels (regression); all statistics are first computed within each seed and then aggregated across the five seeds, and class or bin pairs are not treated as

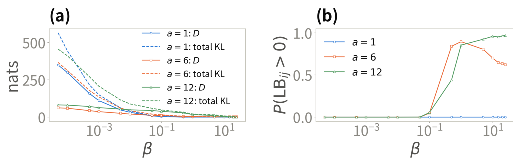

Figure 1: Direct mismatch diagnostics over the ImageNet-100 compression sweep. (a) Anchor mismatch $\widehat{D}$ (solid) and total KL (dashed) against β, by anchor scale. (b) Fraction of class pairs with a positive separation lower bound. Means are over five seeds; no checkpoint is omitted.

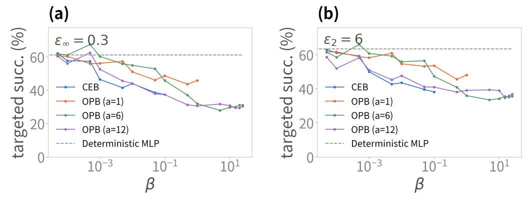

Figure 2: Targeted white-box PGD on MNIST (MLP). (a) L_∞, $\varepsilon = 0.3$; (b) L_2, $\varepsilon = 6$. Curves use the stochastic path and are shown only where clean accuracy exceeds 90%. Transfer curves are in Appendix B.5.

independent replicates. For test examples of class k, define

$$\widehat{D}_k := \frac{1}{n_k} \sum_{n: y_n = k} \frac{\|\mu_\phi(x_n) - a q_{\theta,k}\|^2}{2\widehat{\tau}^2_{\max}}, \qquad \widehat{m}_k := \frac{1}{n_k} \sum_{n: y_n = k} \mu_\phi(x_n). \tag{20}$$

Jensen gives the per-class check $\|\widehat{m}_k - a q_{\theta,k}\| \le \sqrt{2\widehat{\tau}^2_{\max} \widehat{D}_k}$, and the triangle inequality gives the empirical separation bound

$$\widehat{\mathrm{LB}}_{ij} = \sqrt{2}a - \sqrt{2\widehat{\tau}^2_{\max} \widehat{D}_i} - \sqrt{2\widehat{\tau}^2_{\max} \widehat{D}_j}, \tag{21}$$

evaluated without clipping. Across all OPB scales, β values, seeds, and classes, no Jensen or separation-slack violation exceeds 10^{-6}. CEB's mismatch is shown descriptively, but its bound and slack entries are left empty because its reference has no declared Δ.

On ImageNet-100, the two panels read the condition and its certified consequence together (Fig. 1). Mismatch falls with compression: from 62.3 nats at $\beta = 5 \cdot 10^{-5}$ to 2.54 at $\beta = 25$ for $a = 6$, with the total KL co-plotted, and the ordering across scales reverses over the same range. At weak compression the smallest frame is farthest from its anchors (349.8, against 82.1 and 62.3), while at $\beta = 25$ the $a = 12$ frame is tightest among the separated frames (0.56 against 2.54): the anchor-energy readout's prediction optimum sits exactly at the anchors, so a larger declared scale pulls harder and the tied readout is self-enforcing. The positive-bound fraction rises exactly where the corrections shrink: $a = 12$ climbs to 96.9% at $\beta = 25$, $a = 6$ peaks at 89.5% near $\beta = 1$ and settles at 62.3%, and $a = 1$ never certifies, its corrections exceeding the declared margin at every β as its centers collapse under strong compression. The bound and slack readings, the decoupling scatter, and the regression panels are collected in Appendix A.6.

3.5 Adversarial robustness

We use the targeted MNIST PGD protocol of CEB with EOT-10 gradients and 10-sample stochastic predictions; each curve is shown only where the stochastic clean accuracy exceeds 90% (Fig. 2). Within that valid range, targeted success falls monotonically with compression: every curve starts at the deterministic baseline's level ($\approx 61\%$ under L_∞, $\approx 63\%$ under L_2) and declines by roughly a third to a half by the end of its valid range. The models differ in how far that range extends. CEB's curve ends at $\beta = 0.1$, where its success (37.4%/38.1%) is matched by OPB $a = 12$ (37.4%/41.0%); beyond it CEB collapses toward chance accuracy, where near-zero success is trivial rather than a robustness gain, so no later point is plotted. OPB $a = 6, 12$ stay valid over the whole grid and continue to fall, reaching $31.1\%/30.5\%$ under L_∞ and 35.7%/36.8% under L_2 at $\beta = 25$, roughly 30 points below the deterministic baseline under L_∞ and about 27 under L_2, with clean accuracy still $\approx 98\%$; the curves plateau from $\beta \ge 5$ onward, so the gain sits in the compressed operating point rather than in β alone. OPB $a = 1$ stops at $\beta = 1$, where its sampled-path accuracy drops below 90% as the posterior variance relaxes to the prior level (the same effect as in Section 3.8). Transfer attacks are reserved for the gradient-masking check in Appendix B.5.

variant	MNIST (Acc %)	ImageNet-100 (Acc %)	Housing (R^2)
Base	$98.3_{\pm 0.1}$	$83.1_{\pm 0.1}$	$0.762_{\pm 0.024}$
NCM-learn	$98.2_{\pm 0.2}$	$82.2_{\pm 0.4}$	–
NCM-ortho	$98.1_{\pm 0.1}$ ($a = 1$)	$82.3_{\pm 0.6}$ ($a = 1$)	–
CEB	$98.4_{\pm 0.1}$ ($\beta = 0.01$)	$82.8_{\pm 0.2}$ ($\beta = 0.01$)	$0.789_{\pm 0.002}$ ($\beta = 0.0001$)
CEB-energy	$98.2_{\pm 0.1}$ ($\beta = 0.001$)	$81.6_{\pm 0.2}$ ($\beta = 0.001$)	–
GPB-L	$98.4_{\pm 0.1}$ ($\beta = 0.1, a = 12$)	$83.4_{\pm 0.7}$ ($\beta = 1, a = 12$)	–
GPB	$98.4_{\pm 0.1}$ ($\beta = 0.1, a = 12$)	$84.2_{\pm 0.1}$ ($\beta = 1, a = 12$)	–
OPB-FS	$98.5_{\pm 0.0}$ ($\beta = 0.1, a = 12$)	$84.3_{\pm 0.2}$ ($\beta = 1, a = 12$)	–
CEB-tied	–	–	$0.781_{\pm 0.006}$ ($\beta = 0.01$)
EPB-L	–	–	$0.791_{\pm 0.003}$ ($\beta = 0.0001, a = 6$)
EPB	–	–	$0.788_{\pm 0.004}$ ($\beta = 0.0001, a = 6$)
EPB-FS	–	–	$0.787_{\pm 0.008}$ ($\beta = 0.001, a = 6$)

Table 3: Attribution ablation: best-configuration test metrics (mean ± std over five seeds). Configurations are selected by the mean test metric, applied identically to every variant, so all comparisons remain paired by seed; the selection is slightly optimistic in level but shared across the table. One OPB-FS configuration (ImageNet-100, $\beta = 10^{-4}$, $a = 1$) diverged during training (free scales grow without penalty at negligible β) and is excluded.

3.6 Factor-wise attribution

The main comparison changes several factors simultaneously, so we also report a paired intervention ladder in a common operating environment: every variant is re-trained at its best shared configuration on the three settings of the mechanism analysis (Table 3), and all comparisons are paired by seed. On ImageNet-100, GPB reaches 84.2 points, within 0.1 of the free-scale OPB-FS and well above CEB (82.8); on MNIST all variants tie within 0.4 points, and on Housing every EPB variant ties within 0.01 R^2, consistent with the flat strong-compression plateaus of Section 3.7.

The ladder switches each factor independently. *NCM-learn* removes the bottleneck entirely: deterministic $z = \mu_\phi(x)$, energy readout on trainable prototypes, no posterior and no KL. *NCM-ortho* fixes the prototypes to the orthogonal frame $a\, e_k$: fixed geometry, still no IB. *CEB-energy* and *CEB-tied* mount the same energy readout (resp. tied projection) on CEB's trainable free reference. *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable, so only the fixed scale is switched off. Together with Base, CEB, GPB-L and GPB this isolates geometry (CEB-energy vs. OPB-FS), scale (OPB-FS vs. GPB), readout (GPB-L vs. GPB; CEB vs. CEB-energy), and the IB term itself (NCM-ortho vs. GPB).

The paired contrasts (accuracy points on ImageNet-100, 95% CIs in Appendix B.2) rank the factors: geometry is the largest ($+2.67$, OPB-FS over CEB-energy), followed by the IB term ($+1.89$, GPB over NCM-ortho, which additionally contains frame trainability) and the tied readout ($+0.82$, GPB over GPB-L), while fixing the anchor scale costs essentially nothing (-0.04, GPB over OPB-FS). Two patterns matter as much as the ranking. First, the readout contrast flips sign with the accompanying geometry: on the free-geometry side the energy readout costs 1.19 points (CEB-energy below CEB), so restricting the readout helps only when the geometry it ties to is the constructed one. Second, the prototype readout alone does not work: NCM-learn trails the plain backbone by 0.96 points, so the energy form is not the source of the gain either. On MNIST all contrasts shrink to within 0.35 points, and on Housing to within 0.01 R^2, so the ladder resolves factors only where the task margin is large enough. These interventions change the fitted training problem and therefore support factor-wise attribution, not a unique causal decomposition.

3.7 Compression diagnostics

For compactness, the main text shows the classification sweep on ImageNet-100 (MLP); the companion California Housing regression sweep is in Appendix B.3. The larger OPB anchors remain on a high-accuracy plateau farther into strong compression, while CEB and $a = 1$ can collapse; larger anchors also incur a weak-compression cost. The curves make the anchor-scale/compression trade-off visible across the useful operating range.

At $\beta = 25$, OPB with $a = 12$ remains at 79.4% accuracy while CEB has already left its valid high-accuracy range; at weak compression the fixed-scale head pays a one-to-three point cost. The realized KL panel shows where the two objectives spend their penalty: CEB's KL collapses as its learned reference shrinks to absorb the term (3.6 at $\beta = 0.1$ to $\sim 10^{-5}$ at $\beta = 0.5$), whereas OPB's

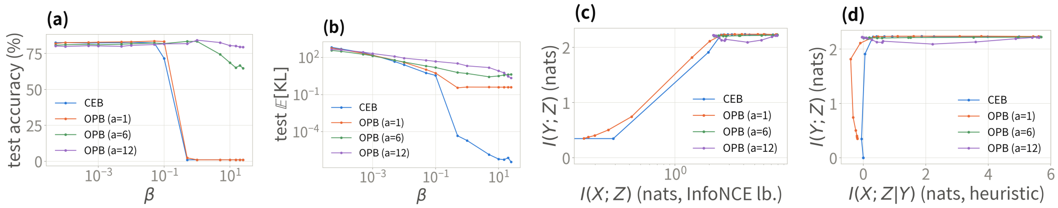

Figure 3: Compression diagnostics across two views and two tasks. (a)–(b) ImageNet-100 classification sweep (MLP): test accuracy and realized conditional KL against β; CEB and the smallest anchor collapse at strong compression, whereas $a = 12$ retains the plateau. The Housing regression counterpart is provided in Appendix B.3. (c)–(d) MNIST information-plane and certificate-plane estimates (estimator-level diagnostics rather than mutual-information identities; Appendix B.4).

Model	Geometry concentration	Scale following	Tied metric	Cross-seed stability
CEB (MNIST)	0.1205 ± 0.0026	5.14	98.16%	0.0201
OPB (MNIST)	0.0203 ± 0.0021	11.00	98.44%	0.0082
CEB (Housing)	0.0233	2.65 ± 0.59	$R^2 = 0.491$	0.593
EPB (Housing)	0.0002	9.22 ± 0.15	$R^2 = 0.778$	0.155

Table 4: Compact mechanism snapshot at matched compression (cross-seed stability is the standard deviation over the five runs of the geometric concentration quantity). Full definitions are in the stored evaluation report.

KL stays anchored at the geometric scale of its fixed prior ($a = 12$: 48 at $\beta = 0.1$, 21 at $\beta = 1$) and is dominated by the mean-mismatch term, so the anchor structure is retained rather than absorbed.

3.8 Information-plane diagnostic

The MNIST information-plane estimates provide a second view of the compression behavior (Fig. 3c,d). They are estimator-level diagnostics rather than mutual-information identities; the matched table and estimator details are in Appendix B.4, with the Housing companion in Appendix A.9.

The two planes show what each method discards as compression deepens. In the information plane (Fig. 3c, log $I(X;Z)$ axis), CEB follows the classic corner: compression and task information fall together, from $I(X;Z) = 2.53$ nat at $\beta = 0.1$ to 1.96 nat at $\beta = 0.5$, and by $\beta = 1$ both are near zero. OPB with $a = 6, 12$ compresses from ≈ 7.9 to ≈ 2.2 nat but never below it, with $I(Y;Z)$ flat at the $H(Y) = \ln 10 \approx 2.30$ ceiling, so compression stops at the task-information level. $a = 1$ tracks $a = 6, 12$ until $\beta = 0.5$ and then collapses toward CEB's corner, with $I(Y;Z)$ falling to 0.35–0.74 nat (Appendix B.4). The certificate plane (Fig. 3d) reads the same contrast as where each trajectory ends: CEB crosses the zero-residual line and $I(Y;Z)$ collapses with it, whereas $a = 6, 12$ terminate on it within ± 0.2 nat of zero residual at full task height. Negative residuals are estimation error rather than information.

3.9 Structural diagnostics

At matched compression $(\beta, a, \rho) = (0.1, 12, 12)$, the posterior geometry is read directly off the class-center cosine Gram matrices (Fig. 4a,b): OPB's Gram is visually an identity matrix, with off-diagonal mean 0.0203, whereas CEB's shows visible off-diagonal structure (0.1205) with no clean pattern. The posterior thus inherits the constructed orthogonal frame, which is exactly what the zero-parameter tied readout relies on; CEB's free reference leaves its posterior entangled, and its free head must re-absorb that geometry. The tied heads attain 98.44% on MNIST against CEB's 98.16%, and $R^2 = 0.778$ on Housing against 0.491. On Housing (Fig. 4c,d), the axial scatter shows the same contrast: EPB's points hug its prior line (off-axis fraction 0.0002), whereas CEB's cloud deviates from its prior line's direction (0.0233) and its learned axis scale is 2.65 ± 0.59 across runs, so its geometry is not a stable structure for a tied readout to exploit.

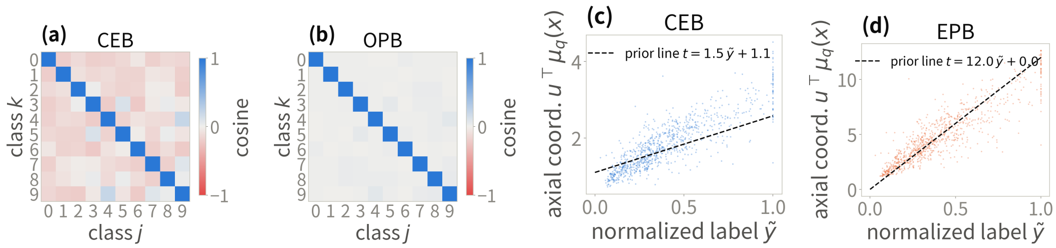

Figure 4: Posterior geometry at the matched operating point: class-center Gram matrices for (a) CEB and (b) OPB, and axial coordinates for (c) CEB and (d) EPB.

Conclusion and limitations. The experiments form a consistent progression from construction to behavior: the baseline table establishes end-to-end utility across modalities; the factor interventions attribute the gain to the geometric prior; the controlled noise study matches the analytic ambiguity floor; and the mismatch, robustness, compression, information-plane, and geometry diagnostics show how the declared prior is reflected in the learned posterior at different operating points. Constraining the conditional prior changes the geometry available to the KL, and the tied readout makes that geometry visible to prediction; the measured mismatch determines how much of the declared geometry transfers, and the anchor scale determines where the trade-off is useful.

The results are qualified accordingly. The mismatch bound is comparator-relative: on a real benchmark its floor is determined by a prespecified comparator and its label, mean-approximation, and covariance terms rather than an intrinsic dataset constant, and the separation lower bound is informative only when the realized mismatch stays below the declared margin, so anchor scale remains an operating choice. The information-plane quantities are estimator-level diagnostics rather than mutual-information identities. Data-driven comparator certificates, adaptive scale selection, and tighter information estimates are natural next steps. Proofs and all supplementary evidence are collected in Appendices A–C.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1; the geometric prior maintains no EMA prototypes or batch statistics and the orthogonalization is exact wherever defined (Assumption A.1). All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix A.8.

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A DEFERRED PROOFS, CONSTRUCTION DETAILS, AND SCOPE

A.1 FAILURE DIAGNOSIS

Proposition A.1 (Dead knob, following Caveat 1 of Kolchinsky & Tracey (2019))**.** *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X;Z|Y) - \gamma I(Y;Z)$ is minimized exactly on the corner family $\{Z : I(Y;Z) = H(Y),\ I(X;Z|Y) = 0\}$, and the map $\gamma \mapsto \big(I(X;Z^*|Y), I(Y;Z^*)\big)$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

Proof sketch (see Kolchinsky & Tracey (2019) for the full argument). By the chain rule, $L_\gamma(Z) = I(X;Z) - (1+\gamma)I(Y;Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper's maximizing convention this is their Eq. 7 with $\beta' = 1/(1+\gamma) \in (0,1)$, the range in which every maximizer is pinned at the copy corner. □

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner *is* the MNI point, so CEB's declared target and the degenerate attractor coincide. The Caveats paper repairs this by squaring the compression term; our anchored reference supplies the same strict convexity by a different mechanism, without squaring: a quadratic displacement term built into the KL by construction (Proposition A.7 below). We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration.

Proposition A.2 (Retention blindness on T_α)**.** *On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky & Tracey (2019) (their Eq. 6), a family that depends on Y alone, $I(X;T_\alpha|Y) = 0$ for every α, and with the optimal backward encoder and classifier, $I(X;T_\alpha|Y) = 0$ and $\mathrm{CE}(T_\alpha) = H(Y) - I(Y;T_\alpha) =: \delta_\alpha$: the certificate reads 0 from the total collapse ($\alpha = 0$) to the full copy ($\alpha = 1$), while the retained label information $I(Y;T_\alpha)$ sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of Y: the certificate records nothing about how much label information the code retains.*

Proof. T_α is a function of Y alone, hence $T_\alpha \perp X \mid Y$ and $I(X;T_\alpha|Y) = 0$. By the decomposition behind Eq. (1), $\mathbb{E}\,\mathrm{KL}(q_\phi(z|X)\|r(z|Y)) = I(X;Z|Y) + \mathbb{E}_Y\,\mathrm{KL}(q_\phi(z|Y)\|r(z|Y))$, with the true backward encoder the gap term vanishes and the certificate is tight, $I(X;T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z)\rangle = -H(Y|T_\alpha)$, i.e. $\mathrm{CE} = H(Y|T_\alpha) = \delta_\alpha$. □

Remark A.1 (The objective on T_α is β-independent and strictly prefers the full copy). On T_α the compression term contributes nothing ($I(X;T_\alpha|Y) \equiv 0$), so the objective reads $L = \mathrm{CE} + \beta\, I(X;T_\alpha|Y) = \delta_\alpha$, independent of β and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits.

Scope of the comparison argument. Theorem 2.1 applies to arbitrary P_{XY} and every feasible comparator in the fitted model class; it does not require $q(z|x) = p_\theta(z|y)$. Its usefulness depends on the comparator KL. Label ambiguity, limited encoder capacity, and covariance misspecification produce the non-zero floor made explicit in Proposition B.1. The stronger statement that this floor is zero still requires the assumptions of Corollary B.2: deterministic labels, exact posterior–prior matching, and a compatible head. We do not assume those exact-realizability premises for the real-data benchmarks. Their performance comparison evaluates utility, while their structural diagnostics report realized mismatch and geometry. The controlled known-noise experiment instead evaluates the ambiguity term where its population value is analytic, including the deterministic distribution as its zero-floor special case.

A.2 PROOFS OF THE COMPARATOR AND SEPARATION GUARANTEES

Proof of Theorem 2.1. Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$ is equivalent to $\Sigma_\theta^{-1}(y) \succeq \tau_{\max}^{-2} I_d$, the Gaussian KL contains a mean term $\frac{1}{2}(\mu_\phi(x) - m_\theta(y))^\top \Sigma_\theta^{-1}(y)(\mu_\phi(x) - m_\theta(y)) \geq \|\mu_\phi(x) - m_\theta(y)\|^2/(2\tau_{\max}^2)$.

Hence $\mathcal{D}(\widehat{w}) \leq \mathcal{K}(\widehat{w})$. For any feasible w_0 in the same family, ε-optimality of $\widehat{w}$, $\ell \geq 0$, and $\beta > 0$ give

$$\beta\mathcal{K}(\widehat{w}) \leq \mathcal{R}(\widehat{w}) + \beta\mathcal{K}(\widehat{w}) \leq \mathcal{R}(w_0) + \beta\mathcal{K}(w_0) + \varepsilon.$$

Division by β proves Eq. (12). □

Proof of Theorem 2.2. Jensen's inequality gives $\|m(y) - m_\theta(y)\| \leq \sqrt{2\tau_{\max}^2 D(y)}$ (the norm is convex, and $\mathbb{E}\|\mu_\phi(X)\|^2 < \infty$ makes $m(y)$ well defined), and the triangle inequality on $m(y) - m(y') = (m(y) - m_\theta(y)) + (m_\theta(y) - m_\theta(y')) + (m_\theta(y') - m(y'))$ gives the two-level bound of Eq. (14), the last step by the definition of $\Delta(y, y')$. The first level is positive exactly when $\Gamma_{yy'} < 1$ and the declared-margin level exactly when $\sqrt{2\tau_{\max}^2 D(y)} + \sqrt{2\tau_{\max}^2 D(y')} < \Delta(y, y')$; for OPB the anchor pair distance is the constant $\sqrt{2}a$ of Eq. (7), so the declared bound satisfies $\Delta(y, y') = \sqrt{2}a$ and the two conditions coincide. With equal mismatch D the common condition $\Gamma_{yy'} = 2\sqrt{2\tau_{\max}^2 D}/(\sqrt{2}a) < 1$ reduces to $D < a^2/(4\tau_{\max}^2)$. For continuous labels the Jensen inequality holds pointwise for P_Y-a.e. y (regular conditional expectation), and intersecting the two measure-one sets of valid y's yields the $P_Y \otimes P_Y$-almost-everywhere pair statement. □

Remark A.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of the continuous case are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max}\, \mathrm{diam}(B_i) + \rho_{\max}\, \mathrm{diam}(B_j)$ (by the upper bound of Eq. (4)), which vanishes as the bins shrink. The separation statement itself is pointwise and needs no binning.

Proof of Corollary 2.3. Theorem 2.1 gives $\mathcal{D}(\widehat{w}_\beta) \leq B_\beta$. For discrete labels, the tower property gives $\mathcal{D}(\widehat{w}_\beta) = \sum_k \pi_k D(k)$, hence $D(k) \leq B_\beta/\pi_k \leq B_\beta/\pi_{\min}$; for continuous labels, Markov's inequality on the nonnegative random variable $D(Y)$ gives $P_Y(D(Y) > B_\beta/\xi) \leq \xi$. In both cases Theorem 2.2 then gives Eq. (17). The rate statement follows directly from the definition of B_β: with $\mathcal{K}(w_0) = 0$ and $\sup_\beta \varepsilon_\beta < \infty$, $B_\beta = O(\beta^{-1})$ so the certified deficit is $O(\beta^{-1/2})$; with $\mathcal{K}(w_0) > 0$, $B_\beta \to \mathcal{K}(w_0)$ as $\beta \to \infty$. □

Proof of Proposition B.1. For OPB let $A_Y := aq_{0,Y}$. Conditional on X, orthonormality gives $\mathbb{E}[A_Y \mid X] = a\sum_k \eta_k(X) q_{0,k}$ and

$$\mathbb{E}\left[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X\right] = a^2\left(1 - \|\eta(X)\|_2^2\right).$$

The conditional bias–variance identity then yields

$$\mathbb{E}\|h(X) - A_Y\|^2 = a^2\mathbb{E}[1 - \|\eta(X)\|_2^2] + \mathbb{E}\left\|h(X) - a\sum_k \eta_k(X) q_{0,k}\right\|^2.$$

Adding the diagonal-Gaussian covariance term V_0 proves Eq. (27). For EPB set $A_Y := \rho\tilde{Y}u_0$. Then $\mathbb{E}[A_Y \mid X] = \rho\nu(X)u_0$ and $\mathbb{E}[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X] = \rho^2\,\mathrm{Var}(\tilde{Y} \mid X)$. The same identity plus V_0 proves Eq. (28). □

Proof of Corollary B.2. For classification, the assumed comparator $q(z|x) = p_\theta(z|y)$ has $\mathcal{K}(w_0) = 0$. Its Bayes cross-entropy is $C_{\mathrm{G}} = H(Y \mid Z_{\mathrm{align}}) \leq \ln K$; with shared covariance the determinant factors cancel and its posterior is the anchor-energy rule. Applying Eq. (12) gives Eq. (29). For regression the aligned comparator again has zero KL. Under the tied head its error is $(\tau/\rho)\eta$, $\eta \sim \mathcal{N}(0, 1)$, so its standardized squared risk is τ^2/ρ^2; the same proposition gives Eq. (30). □

Proof of Proposition B.3. By Jensen's inequality the left side is at least $\frac{1}{2\tau^2}\sum_k \pi_k\|c - aq_{\theta,k}\|^2 = \frac{1}{2\tau^2}\left(\|c\|^2 - 2a\,c^\top\sum_k \pi_k q_{\theta,k} + a^2\right)$. Minimizing over c gives $c = a\sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding Eq. (31). □

Remark A.3 (What the framework constrains (and what it does not)). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y = k) = \mathcal{N}(a\,q'_k, \tau^2 I_d)$, the trainable frame can rotate to match ($q_{\theta,k} \to q'_k$), yielding $\mathrm{KL} = 0$, $I(X; Z|Y) = 0$, and $I(Y; Z) > 0$: a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. In the fixed-covariance canonical instance, the remaining freedom at certificate zero is the frame rotation. The covariance-relaxed real-data protocol has an additional but log-clamped matching channel; the comparison theorem applies with the resulting $\tau^2_{\max}$, and Section 3.4 reports that channel rather than treating it as fixed.

A.3 CONSTRUCTION DETAILS

Assumption A.1 (Full rank)**.** *U_θ has full column rank wherever the polar factor and its backward pass are evaluated; this is monitored rather than imposed by an extra loss.*

Polar permutation equivariance. For any label permutation matrix P,

$$Q_\theta(U_\theta P) = U_\theta P\big(P^\top U_\theta^\top U_\theta P\big)^{-1/2} = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2} P = Q_\theta(U_\theta)\, P, \tag{22}$$

since $PP^\top = I_K$ and the inverse square root conjugates under P. Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

Proposition A.3 (Orthogonal class-prior geometry by construction)**.** *Under Eqs. (6)–(7), the class means $a_k = a\, q_{\theta,k}$ have Gram matrix $a^2 I_K$: under Assumption A.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.*

Proof. The geometry in Eq. (7) follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by a; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. □

Proposition A.4 (Anchor-energy classifier equivalence)**.** *The scores $s_k(z) = -\|z - aq_{\theta,k}\|^2/(2\tau^2) + \log \pi_k$ are Gaussian Bayes scores up to a class-independent term. For uniform classes, the corresponding linear weights are $(a/\tau^2)q_{\theta,k}$.*

Initialization, monitoring, and re-initialization. The posterior logvar head is zero-initialized ($\sigma^2 = \tau^2 = 1$ at initialization, so training starts at KL $\approx \frac{1}{2\tau^2}a^2$); the prior encoder is identity-initialized, so the initial raw matrix is $U_\theta = [e_1, \ldots, e_K]$, full column rank, with a stable polar-factor backward (the verified condition of Assumption A.1). The full-rank condition is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. (5) exactly. Near rank deficiency the polar factor's gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

Covariance protocol. Equations (7) and (9) define the fixed-covariance canonical instances used for the exact Bayes-head equivalence and the controlled label-ambiguity experiment of Section 3.3. The real-data benchmark protocol retains the common CEB parameterization of a learned label-conditional diagonal covariance, with log-variance clamped to a fixed interval. It is therefore a covariance-relaxed member of the geometric prior family $\mathcal{G}$: the OPB/EPB mean geometry and readout scale remain fixed, while Eq. (2) holds with the clamp-induced $\tau^2_{\max}$. Proposition A.4 is an exact Gaussian Bayes equivalence only for the shared-covariance canonical instance; in the relaxed benchmark it remains a zero-parameter readout of the same anchor means. The direct diagnostics in Section 3.4 use each checkpoint's realized conditional covariance rather than silently substituting the canonical $\tau^2 I$.

EPB axis choices. Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \ldots, 0]$, with no trainable prior direction: the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with W trained and normalized at every step, our main variant, which keeps the scale ρ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$: the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

EPB isometry notes. The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to ρ; unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with $|\tilde{y}|$; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for $y_i < y_j < y_k$ the distances add, $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$: order and magnitude of label differences are encoded exactly.

Multidimensional regression labels. For $y \in \mathbb{R}^m$ ($m \leq d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d\times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho\, Q\, \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise "isometric" has no unique meaning.

A.4 Additional results

Lemma A.5 (OPB KL decomposition)**.** *For a diagonal posterior covariance and the prior of Eq. (7),*

$$\mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big) = \frac{1}{2\tau^2}\|\mu_\phi(x) - a\,q_{\theta,y}\|^2 + \frac{1}{2}\sum_{r=1}^{d}\Big[\frac{\sigma^2_{\phi,r}(x)}{\tau^2} - 1 - \log\frac{\sigma^2_{\phi,r}(x)}{\tau^2}\Big], \quad (23)$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma^2_{\phi,r}(x) = \tau^2$ for all r. The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.7 below.

Proof. Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a\,q_{\theta,y}$, posterior variances $\sigma^2_{\phi,r}(x)$, and the fixed prior variance τ^2. ☐

Corollary A.6 (Label-information lower bound in the aligned Gaussian model)**.** *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a\,q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\mathrm{pair}} = \Phi\big(-\frac{a}{\sqrt{2}\,\tau}\big)$, with $P_e \leq \min\{1, (K-1)P_{\mathrm{pair}}\}$, and whenever the displayed bound $\bar{P}_e \leq 1-1/K$, Fano's inequality gives $I(Y;Z) \geq \log K - h_2(\bar{P}_e) - \bar{P}_e\log(K-1)$. This links the anchor scale-to-noise ratio a/τ to the retained label information, conditionally on the aligned Gaussian model.*

Proof. Two anchors have distance $\sqrt{2}a$ by Proposition A.3; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano's inequality on $[0, 1 - 1/K]$ conclude. ☐

Proposition A.7 (Joint strict convexity and unique optimum on a restricted aligned surrogate)**.** *Consider the class-symmetric aligned family: every input of class y is mapped to the posterior $q_y = \mathcal{N}(s\,q_{\theta,y}, \sigma^2 I)$ ($s \geq 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier of Proposition A.4 (weights $w_j = c\,q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the* stochastic training code *$z \sim q_y$ (as in the actual objective, Eq. (5)), the training cross-entropy is*

$$g(s,\sigma^2) := \mathbb{E}_\eta\Big[\ln\Big(1 + \textstyle\sum_{j\neq y} e^{-cs + c\sigma(\eta_j - \eta_y)}\Big)\Big], \qquad \eta \sim \mathcal{N}(0, I_K), \quad (24)$$

and the training Lagrangian is

$$L_\beta(s,\sigma^2) = g(s,\sigma^2) + \beta\Big[\tfrac{1}{2\tau^2}(s-a)^2 + \tfrac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)\Big]. \quad (25)$$

Then for every $\beta > 0$, L_β is strictly convex jointly in (s,σ) and has a unique joint minimizer $(s^*(\beta), \sigma^*(\beta))$, interior in s since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every β (Proposition A.1). This is a restricted surrogate result*: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective, with its nonlinear posterior, trainable frame, and non-convex degrees* *of freedom, lies outside this analysis, and the result neither establishes β-tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ($z = \mu$) is the $\sigma \to 0$ limit, and $g \geq \ln(1 + (K-1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do* not *claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer σ^* responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$, and establishing the sign of $\mathrm{d}s^*/\mathrm{d}\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the β-sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition A.4), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution: at $\mathrm{KL} \to 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$ it contains $\mathbb{E}[\ln(1 + e^{-c(a+\tau\xi)})]$, whose integrand grows linearly in c on the event $\xi < -a/\tau$, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.*

Proof. For each fixed η, the integrand of Eq. (24) is a log-sum-exp of functions affine in (s, σ) (the constant 1 has slope zero and each exponential has slope $(-c,\, c(\eta_j - \eta_y))$), hence jointly convex in (s, σ), and the expectation g is jointly convex. No strictness is claimed for g: for $K = 2$ the integrand depends on (s, σ) only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s, \sigma) := \frac{1}{2\tau^2}(s-a)^2 + \frac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)$; its Hessian in (s, σ) is $\nabla^2 R = \mathrm{diag}\big(\tau^{-2},\; d\,(\tau^{-2} + \sigma^{-2})\big)$, strictly positive definite for every $\sigma > 0$, so βR is strictly convex in (s, σ). Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in (s, σ). Since $R \to +\infty$ as $s \to \infty$, $\sigma \to 0^+$, or $\sigma \to \infty$, the sublevel sets of L_β are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in s: at $s = 0$, $\partial L_\beta/\partial s = -c\,\mathbb{E}[p_{\mathrm{w}}] - \beta a/\tau^2 < 0$, where p_{w} is the error probability of the sampled code, so $s^* > 0$. □

Remark A.4 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.7 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale: a restricted surrogate result, which does not establish β-tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB's: Eq. (1) is the usual CEB bound, here for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption A.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified, not imposed, along the reported runs (Section 2.1); the construction is exactly equivariant under relabeling of the classes by Eq. (22), so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and the certificate-preservation argument would no longer follow directly. Finally, the anchor-energy classifier of Proposition A.4 fixes the classifier scale to a/τ^2 by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.7; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.7 moves monotonically in β is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate's zero set contains the full-retention orthogonal label codes at any frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code*; it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

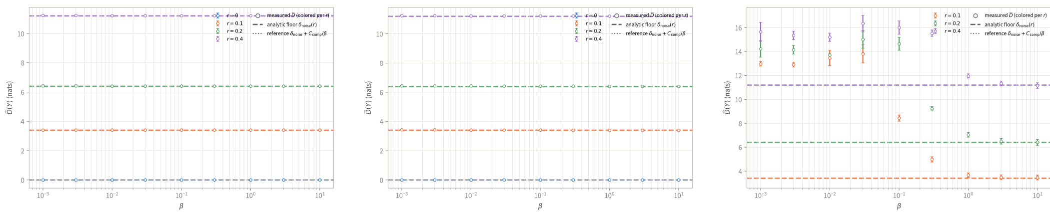

Figure 5: Sensitivity analyses for the controlled label-ambiguity experiment. (a) Posterior variance fixed to the prior variance. (b) Trainable polar frame. (c) Labels sampled rather than integrated with their exact population weights; failed comparator-gate configurations are marked. Dashed lines denote the analytic population ambiguity floors. Points are means and error bars are standard deviations over five seeds.

A.5 Controlled label-ambiguity sensitivity analyses

Training evaluates the population objective by weighting all K^2 clean/observed-label pairs by the noise model, with a descending-β warm start to remove avoidable optimization error at the small-β end; the comparator risk $C_{\mathrm{comp}}(r)$ is estimated by Monte Carlo. The population-weighted experiment of Section 3.3 uses a fixed frame and the standard trainable posterior variance. Figure 5 changes these choices one at a time. Fixing the posterior variance to $\sigma^2 \equiv \tau^2$ leaves the measured mismatch on the analytic floors, confirming that the result is not produced by a covariance compensation channel. Replacing the fixed frame by the trainable polar frame gives the same result, so the floor is invariant to the learned rigid frame rotation.

The third panel replaces the exact population weights by sampled noisy labels. It shows the expected finite-sample displacement from the population floor at weak compression and convergence toward the empirical floor as compression strengthens. Six sampled-label configurations at large β fail the explicit comparator-objective gate (twelve stored rows because both best and final checkpoints are retained); they are marked as optimization failures rather than interpreted as violations of Eq. (12). The population-weighted main grid and both structural controls satisfy the gate at every point.

A.6 Direct mismatch definitions and separation diagnostics

This section collects the diagnostic panels deferred from Section 3.4: the decoupling scatter, the separation-bound and slack readings, and the regression diagnostics, together with the finite-bin definitions. All statistics are first computed within each seed and then aggregated across the five seeds; class or bin pairs are not treated as independent replicates. Every checkpoint in both requested grids is present, and no collapse or variance-explosion point is removed.

The deferred classification panels complete the picture (Fig. 6). The decoupling scatter (a) shows MC-10 accuracy staying within 82–86% while $\widehat{D}$ spans three orders of magnitude: mechanism and task metric are decoupled. Panel (b) shows the raw bound rising toward the declared $\sqrt{2a}$ exactly where the mismatch corrections shrink, so the slack narrows where the bound is informative; CEB's mismatch is plotted descriptively, without a bound, since its reference has no declared Δ. The slack percentiles (c) stay non-negative at every reported point: the bound is consistent but conservative.

Regression. We partition the standardized test labels into 20 equal-width bins and merge any bin containing fewer than 30 examples. Let $\bar{y}_b$ be the mean label of bin B_b, $\widehat{m}_b$ its posterior center, $\widehat{D}_b$ its average sample-to-own-anchor mismatch, and $h_b := \mathrm{diam}(B_b)$. The finite-bin lower bound is

$$\widehat{\mathrm{LB}}_{bc} = \rho|\bar{y}_b - \bar{y}_c| - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_b} - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_c} - \rho(h_b + h_c), \tag{26}$$

where the last term is the quantization correction of Remark A.2. On Housing with $\rho = 6$, mismatch falls from 0.264 to 0.026, the axial slope remains 0.73–0.77, and the mismatch is almost entirely axial (about 0.2 against 0.004 off-axis), while R^2 changes only from 0.786 to 0.767 over the sweep: a tenfold mismatch reduction need not imply a comparable task-metric gain. Figure 6d plots the actual bin-center distances against the declared isometric distances at $\rho = 6$: the cloud hugs a line of slope 0.73–0.77 below the identity line at all four β values, so the posterior follows the declared

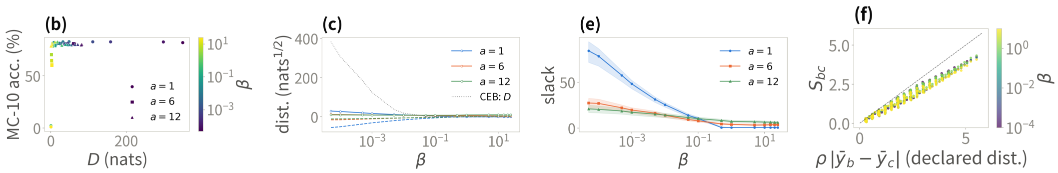

Figure 6: Direct mismatch diagnostics deferred from the main text. (a) MC-10 accuracy against $\widehat{D}$ (color $=\beta$, marker $=a$). (b) Mean separation (solid) and raw lower bound (dashed) against β; gray dotted: CEB, descriptive. (c) Slack percentiles. (d) Cal. Housing bin-pair separation against the declared prior distance at $\rho=6$ (color $=\beta$; identity line: exact transfer). Means and standard deviations are over five seeds; no checkpoint is omitted.

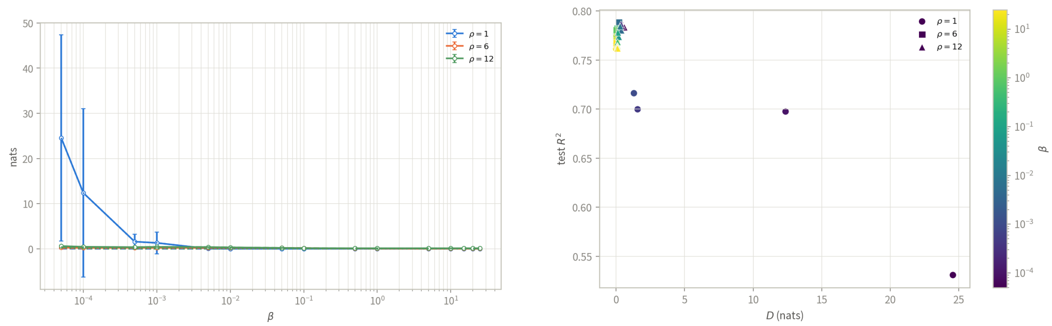

Figure 7: Cal. Housing mismatch sweep panels (deferred from the main text). (a) Anchor mismatch $\widehat{D}$ against β, by isometric scale ρ, decomposed into axial and off-axis contributions. (b) Test R^2 against $\widehat{D}$; color denotes β and marker shape ρ. Means and standard deviations are over five seeds; no checkpoint is omitted.

metric structure at about three quarters of its scale, and the scale-transfer ratio is essentially β-invariant. After the quantization correction, 43% of bin pairs retain a positive bound. Figure 7 shows the two regression sweep panels: the mismatch–β curves with their axial/off-axis decomposition (the dotted off-axis contribution stays below 0.01 nats, so the posterior lies essentially on the declared axis), and the R^2-against-$\widehat{D}$ scatter whose band stays within 0.75–0.79 while $\widehat{D}$ spans orders of magnitude.

Normalization and variance extremes. The empirical mismatch uses one maximum conditional-reference variance per checkpoint, whereas the exact Gaussian KL uses every sample's and coordinate's own variance. They therefore need not coincide: averaged over the ImageNet-100 grid, $\widehat{D}=41.2$, the KL mean term is 46.5, and the KL variance term is 51.4. On Housing, a small number of weak-compression, large-ρ variance explosions raise the across-grid mean KL variance term to 2.8×10^6. These checkpoints remain in all plots. Because their $\widehat{\tau}^2_{\max}$ is also large, the normalized $\widehat{D}$ can fall even as the KL variance term grows; the unnormalized axial/off-axis components and total KL are therefore reported alongside it.

A.7 Main results with standard deviations

Tables 5 and 6 report the same configurations as the main table with standard deviations over runs, split by column groups for width (the baselines table includes NCBD and the variants table includes AdaCap); bold and underline follow the same mean-based rule as Table 1.

A.8 Paired-difference confidence intervals

Tables 7–10 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap 95% confidence interval: the differences are paired across the shared

Table 5: Test accuracy (%) and test R^2 with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 1.

Dataset	Backbone	Base	VIB	SVIB	NIB	NCBD
MNIST	MLP	98.33±0.12	98.28±0.11	98.15±0.13	98.27±0.07	97.95±0.23
MNIST	CNN	99.13±0.05	99.28±0.09	99.23±0.03	99.22±0.06	99.18±0.07
ImageNet-100	MLP	83.12±0.14	82.84±0.44	83.02±0.37	82.35±0.27	83.64±0.07
ImageNet-100	CNN	83.40±0.13	82.94±0.37	83.11±0.86	82.91±0.35	80.56±1.86
Cora	GCN	87.68±0.46	86.83±0.28	86.46±1.87	86.35±0.68	87.34±1.60
IMDb	RNN	84.85±0.68	85.50±0.91	85.99±0.30	86.14±0.92	86.69±0.52
AG News	RNN	**92.94±0.15**	92.72±0.17	92.86±0.21	92.91±0.22	92.37±0.37
Cal. Housing	MLP	0.7618±0.0238	0.7901±0.0049	0.7897±0.0058	0.7862±0.0035	–
STS-B	RNN	**0.2899±0.0113**	0.2719±0.0164	0.2697±0.0202	0.2417±0.0221	–
ZINC	GCN	0.6348±0.0044	0.6339±0.0033	0.6325±0.0026	0.6178±0.0099	–
AgeDB	MLP	0.2180±0.0409	0.3173±0.0101	0.3385±0.0286	0.2710±0.0235	–
AgeDB	CNN	0.3914±0.0194	0.3342±0.1894	0.3323±0.1870	0.3337±0.1889	–

Table 6: Test accuracy (%) and test R^2 with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 1. GPB and GPB-L are separated from the references by the vertical rule.

Dataset	Backbone	CEB	DVCCA	AdaCap	GPB	GPB-L
MNIST	MLP	98.43±0.08	98.24±0.17	–	**98.54±0.09**	98.43±0.07
MNIST	CNN	99.30±0.05	99.29±0.10	–	**99.32±0.05**	99.31±0.05
ImageNet-100	MLP	82.77±0.22	82.99±0.43	–	**84.40±0.21**	84.01±0.23
ImageNet-100	CNN	82.97±0.67	83.21±0.70	–	**86.08±0.21**	84.88±0.80
Cora	GCN	86.27±0.38	86.90±0.94	–	**88.12±0.55**	87.86±0.87
IMDb	RNN	85.55±0.73	85.69±0.23	–	86.84±0.57	**86.99±0.35**
AG News	RNN	92.83±0.11	92.78±0.15	–	92.87±0.12	92.92±0.14
Cal. Housing	MLP	0.7894±0.0018	0.7878±0.0048	0.7758±0.0023	0.7895±0.0051	**0.7914±0.0034**
STS-B	RNN	0.2668±0.0127	0.2713±0.0218	0.1613±0.0188	0.2762±0.0152	0.2755±0.0134
ZINC	GCN	0.6257±0.0044	0.6294±0.0026	**0.6770±0.0019**	0.6344±0.0031	0.6323±0.0016
AgeDB	MLP	0.3317±0.0251	0.3180±0.0228	0.3482±0.0209	**0.3536±0.0076**	0.3320±0.0080
AgeDB	CNN	0.3882±0.0275	0.3205±0.1811	0.4684±0.0094	**0.4700±0.0072**	0.4388±0.0220

seeds, the interval is the percentile interval of $B = 10{,}000$ resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is $\delta = 0.2$ percentage points of accuracy or $\delta = 0.005$ in R^2. Bold marks a lower bound above zero (significant advantage) and † a lower bound within the margin (tied); the best configuration per method is the same as in the main table. The last table covers the external non-IB baselines, NCBD on the classification settings and AdaCap on the regression settings; the only interval lying entirely below zero is AdaCap on ZINC (-0.043 R^2).

A.9 Information-plane panels for Cal. Housing

Figure 8 completes panels (c)–(d) of Fig. 3 in the main text with the regression setting, under the same β grid, axis conventions, and five runs per point; here $I(Y;Z)$ is the Gaussian approximation $\frac{1}{2}\ln\big(\mathrm{Var}(y)/\mathrm{Var}(y-\hat{y})\big)$, a heuristic rather than a bound. On Cal. Housing this approximation consistently reads above the InfoNCE bound, so the certificate panel reflects the estimator gap rather than the geometry; the information plane remains informative: CEB compresses itself off the task ($R^2 = 0.49$ at $\beta = 0.1$, 0.18 at $\beta = 0.5$) while every EPB curve keeps $R^2 \geq 0.76$ flat out to $\beta = 25$.

B Additional theory and experiments

This appendix collects the detailed material omitted from the nine-page main text. All reported numbers use the same checkpoints and seeds as the main comparison.

Table 7: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin δ (tied).

Dataset	Backbone	Base	VIB
MNIST	MLP	**+0.2 [+0.2, +0.3]**	**+0.3 [+0.2, +0.3]**
MNIST	CNN	**+0.2 [+0.2, +0.2]**	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	**+1.3 [+1.2, +1.4]**	**+1.6 [+1.4, +1.9]**
ImageNet-100	CNN	**+2.7 [+2.5, +2.9]**	**+3.1 [+3.0, +3.4]**
Cora	GCN	+0.4 [-0.1, +1.1]†	**+1.3 [+0.7, +1.7]**
IMDb	RNN	**+2.0 [+1.2, +2.7]**	**+1.3 [+0.6, +2.3]**
AG News	RNN	-0.1 [-0.2, +0.0]†	**+0.2 [+0.1, +0.2]**
Cal. Housing	MLP	**+0.028 [+0.008, +0.048]**	-0.001 [-0.005, +0.003]†
STS-B	RNN	-0.014 [-0.022, -0.006]	+0.004 [-0.017, +0.022]
ZINC	GCN	-0.000 [-0.003, +0.002]†	+0.001 [-0.004, +0.004]†
AgeDB	MLP	**+0.136 [+0.104, +0.167]**	**+0.036 [+0.026, +0.046]**
AgeDB	CNN	**+0.079 [+0.066, +0.091]**	**+0.136 [+0.043, +0.302]**
		8*/3†	8*/3†

Table 8: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/† as in the baselines table.

Dataset	Backbone	SVIB	NIB
MNIST	MLP	**+0.4 [+0.3, +0.5]**	**+0.3 [+0.2, +0.3]**
MNIST	CNN	**+0.1 [+0.1, +0.1]**	**+0.1 [+0.1, +0.2]**
ImageNet-100	MLP	**+1.4 [+1.1, +1.7]**	**+2.1 [+1.8, +2.2]**
ImageNet-100	CNN	**+3.0 [+2.4, +3.9]**	**+3.2 [+2.9, +3.5]**
Cora	GCN	**+1.7 [+0.2, +3.4]**	**+1.8 [+1.1, +2.4]**
IMDb	RNN	**+0.8 [+0.2, +1.4]**	**+0.7 [+0.2, +1.2]**
AG News	RNN	+0.0 [-0.1, +0.1]†	-0.0 [-0.2, +0.1]†
Cal. Housing	MLP	-0.000 [-0.006, +0.004]	+0.003 [-0.001, +0.009]†
STS-B	RNN	+0.006 [-0.021, +0.028]	**+0.035 [+0.011, +0.058]**
ZINC	GCN	+0.002 [-0.001, +0.004]†	**+0.017 [+0.010, +0.026]**
AgeDB	MLP	+0.015 [-0.010, +0.041]	**+0.083 [+0.062, +0.100]**
AgeDB	CNN	**+0.138 [+0.045, +0.303]**	**+0.136 [+0.040, +0.302]**
		7*/2†	10*/2†

B.1 DEFERRED THEORY STATEMENTS

Proposition B.1 (Ambiguity–approximation decomposition)**.** *For OPB classification with $\eta_k(x) = P(Y = k \mid X = x)$ and a comparator mean $h(x)$,*

$$\mathcal{K}(w_0) = \frac{a^2}{2\tau^2}\mathbb{E}[1 - \|\eta(X)\|_2^2] + \frac{1}{2\tau^2}\mathbb{E}\left\|h(X) - a\sum_k \eta_k(X) q_{0,k}\right\|^2 + V_0, \tag{27}$$

where V_0 is the covariance mismatch. For EPB regression,

$$\mathcal{K}(w_0) = \frac{\rho^2}{2\tau^2}\mathbb{E}[\mathrm{Var}(\tilde{Y} \mid X)] + \frac{1}{2\tau^2}\mathbb{E}\|h(X) - \rho\nu(X)u_0\|^2 + V_0. \tag{28}$$

Substituting either decomposition into B_β in Corollary 2.3 separates the comparator-dependent geometry transfer floor into label ambiguity, encoder mean approximation, and covariance mismatch.

Corollary B.2 (Zero-floor exact-realizability special cases)**.** *Under deterministic labels, exact posterior–prior matching, and a compatible Bayes head, $\mathcal{K}(w_0) = 0$ and Theorem 2.1 gives*

$$\mathcal{D}(\widehat{w}) \leq (\ln K + \varepsilon)/\beta \tag{29}$$

for classification. For EPB regression,

$$\mathcal{D}(\widehat{w}) \leq (\tau^2/\rho^2 + \varepsilon)/\beta. \tag{30}$$

Table 9: Paired-difference 95% confidence intervals of GPB over CEB, DVCCA, and the free-head ablation GPB-L (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/† as in the baselines table.

Dataset	Backbone	CEB	DVCCA	GPB-L
MNIST	MLP	**+0.1 [+0.0, +0.2]**	**+0.3 [+0.2, +0.4]**	**+0.1 [+0.0, +0.2]**
MNIST	CNN	+0.0 [-0.0, +0.1]†	+0.0 [-0.1, +0.1]†	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	**+1.6 [+1.5, +1.7]**	**+1.4 [+1.1, +1.7]**	**+0.4 [+0.2, +0.5]**
ImageNet-100	CNN	**+3.1 [+2.4, +3.7]**	**+2.9 [+2.4, +3.5]**	**+1.2 [+0.5, +1.9]**
Cora	GCN	**+1.8 [+1.5, +2.3]**	**+1.2 [+0.4, +1.9]**	+0.3 [-0.5, +1.3]
IMDb	RNN	**+1.3 [+0.3, +2.1]**	**+1.1 [+0.8, +1.5]**	-0.2 [-0.6, +0.5]
AG News	RNN	+0.0 [-0.1, +0.2]†	+0.1 [-0.0, +0.2]†	-0.0 [-0.2, +0.1]†
Cal. Housing	MLP	+0.000 [-0.003, +0.004]†	+0.002 [-0.005, +0.009]	-0.002 [-0.005, +0.002]
STS-B	RNN	+0.009 [-0.009, +0.025]	+0.005 [-0.020, +0.026]	+0.001 [-0.019, +0.017]
ZINC	GCN	**+0.009 [+0.005, +0.011]**	**+0.005 [+0.002, +0.007]**	+0.002 [-0.001, +0.005]†
AgeDB	MLP	**+0.022 [+0.006, +0.038]**	**+0.036 [+0.012, +0.057]**	**+0.022 [+0.019, +0.025]**
AgeDB	CNN	**+0.082 [+0.064, +0.099]**	**+0.149 [+0.065, +0.310]**	**+0.031 [+0.018, +0.046]**
		8*/3†	8*/2†	5*/3†

Table 10: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in R^2. Bold/† as in the baselines table.

Dataset	Backbone	NCBD	AdaCap
MNIST	MLP	**+0.6 [+0.4, +0.8]**	–
MNIST	CNN	**+0.1 [+0.1, +0.2]**	–
ImageNet-100	MLP	**+0.8 [+0.6, +0.9]**	–
ImageNet-100	CNN	**+5.5 [+4.3, +6.9]**	–
Cora	GCN	**+0.8 [+0.0, +1.9]**	–
IMDb	RNN	+0.2 [-0.5, +0.7]	–
AG News	RNN	**+0.5 [+0.2, +0.8]**	–
Cal. Housing	MLP	–	**+0.014 [+0.009, +0.019]**
STS-B	RNN	–	**+0.115 [+0.086, +0.135]**
ZINC	GCN	–	-0.043 [-0.046, -0.041]
AgeDB	MLP	–	+0.005 [-0.007, +0.021]
AgeDB	CNN	–	+0.002 [-0.004, +0.006]†
		6*/0†	2*/1†

Proposition B.3 (Positive rate cost of class-independent means)**.** *If* $\mathbb{E}[\mu_\phi(X) \mid Y = k] = c$ *for every class, then*

$$\mathbb{E}\left[\frac{\|\mu_\phi(X) - a q_{\theta,Y}\|^2}{2\tau^2}\right] \geq \frac{a^2}{2\tau^2}\left(1 - \sum_k \pi_k^2\right). \tag{31}$$

Corollary B.4 (The orthogonal instance instantiates conditional posterior separation)**.** *For OPB,* $\tau_{\max} = \tau$ *and* $\Delta(y, y') = \sqrt{2}a$*, so*

$$\|m(y) - m(y')\| \geq \sqrt{2}a - \sqrt{2\tau^2 D(y)} - \sqrt{2\tau^2 D(y')}. \tag{32}$$

B.2 Factor-wise ablations

The factor-wise ablation ladder switches geometry, anchor scale, readout, and the IB term separately. The complete means appear in Table 3 of the main text; the paired bootstrap intervals are reported below. Table 11 reports the paired bootstrap intervals for all contrasts; the factor ranking and the sign-flip and negative-result patterns are read in the main text (Section 3.6).

B.3 Regression compression curves

The main text shows the classification compression sweep. This companion figure gives the complete California Housing regression sweep under the same β grid, including the weak-compression

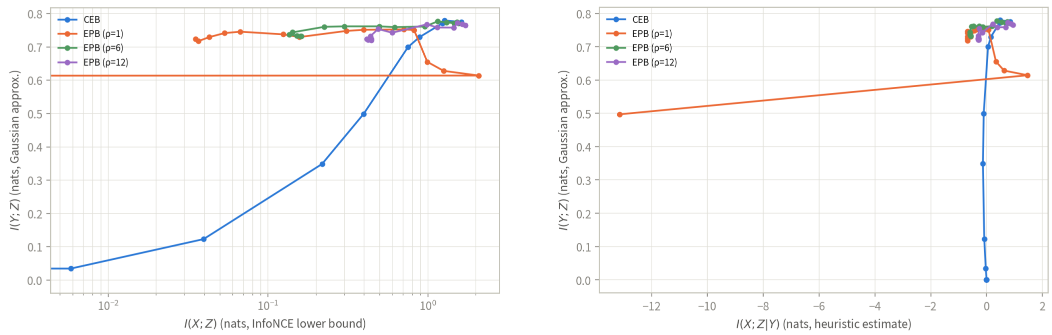

Figure 8: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 3(c,d): identical estimators, β grid (log x-scale in (a); linear where the estimated residual turns negative in (b)), five runs per point, lines connect β in ascending order. CEB one curve per panel; EPB three ($\rho \in \{1, 6, 12\}$).

factor	comparison	MNIST	ImageNet-100	Housing
readout	GPB − GPB-L	$+0.02$ $[-0.05, 0.08]$	$+0.82$ $[0.34, 1.17]$	–
readout (EPB)	EPB − EPB-L	–	–	-0.00 $[-0.01, -0.00]$
readout (free geometry)	CEB-energy − CEB	-0.21 $[-0.30, -0.12]$	-1.19 $[-1.50, -0.96]$	–
geometry	OPB-FS − CEB-energy	$+0.32$ $[0.26, 0.40]$	$+2.67$ $[2.45, 2.92]$	–
geometry (EPB)	EPB-FS − CEB-tied	–	–	$+0.01$ $[-0.00, 0.01]$
fixed scale	GPB − OPB-FS	-0.09 $[-0.16, -0.02]$	-0.04 $[-0.07, -0.01]$	–
fixed scale (EPB)	EPB − EPB-FS	–	–	$+0.00$ $[-0.01, 0.01]$
IB increment*	GPB − NCM-ortho	$+0.34$ $[0.22, 0.45]$	$+1.89$ $[1.41, 2.36]$	–
prototype structure	NCM-ortho − NCM-learn	-0.06 $[-0.20, 0.11]$	$+0.16$ $[-0.49, 0.82]$	–
prototype readout alone	NCM-learn − Base	-0.16 $[-0.28, -0.05]$	-0.96 $[-1.30, -0.69]$	–

Table 11: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100, R^2 on Housing; 95% bootstrap percentile CIs, $B = 10{,}000$, fixed seed, paired across the shared five seeds at each variant's best configuration; positive values favor the first term). *NCM-ortho fixes the frame at $a\, e_k$ while GPB trains it, so the IB comparison additionally contains frame trainability.

variance explosion and the strong-compression EPB plateau. The curves are diagnostics of achieved KL and task performance, not tests of the alignment rate.

B.4 Information-plane estimates

Estimator protocol. $I(X;Z)$ is estimated by the InfoNCE lower bound (van den Oord et al., 2018) ($\log B - \mathcal{L}_{\mathrm{NCE}}$, $B = 4096$; one critic per run, trained on the training split, the bound read off the test split with $z \sim q(z|x)$); $I(Y;Z)$ by the classification lower bound $H(Y) - \mathrm{CE}$; and the residual $I(X;Z|Y)$ is the difference of the two, a heuristic rather than a bound. The true residual is nonnegative by the data-processing inequality; negative plotted values mean the estimators, not the theory, gave way. Because negative residuals occur (OPB $a = 1$; the collapsed CEB trajectory), the certificate-plane axis is drawn linear. The MNIST panels appear in Fig. 3(c,d) of the main text.

Matched-compression performance. Table 12 pairs each CEB point with the OPB configuration of nearest estimated $I(X;Z)$. At ≈ 2.0 nat, CEB ($\beta = 0.5$) retains 86.0% while OPB ($a = 1, \beta = 0.5$) retains 97.8%; at ≈ 2.5 nat ($\beta = 0.1$) the two tie (98.2% vs. 98.1%). These are paired empirical comparisons at two estimated compression levels, not a causal attribution to geometry.

Estimator caveat. OPB ($a = 1$) at $\beta \geq 5$ must be read with care. On the deterministic μ path the accuracy stays 97.6–97.8%, for which Fano's inequality requires $I(Y;Z) \geq H(Y) - h(0.023) - 0.023 \ln 9 \approx 2.1$ nat, so the plotted $I(Y;Z)$ of 0.35–0.74 nat is a bound failure on that path: overconfident errors inflate the cross-entropy, and near-duplicate positives saturate the InfoNCE objective. On the sampled posterior the degradation is partly real: at posterior variance $\tau^2 = 1$ the radius-1 anchors cannot separate classes, and the stochastic-path clean accuracy falls to 0.84 at $\beta = 5$ and 0.59 at $\beta = 25$, against which the CE-based estimate still sits below the Fano floor. The leftward spur therefore mixes a real sampled-path collapse with an estimator failure of the deterministic path; the $a = 6, 12$ readings above are unaffected.

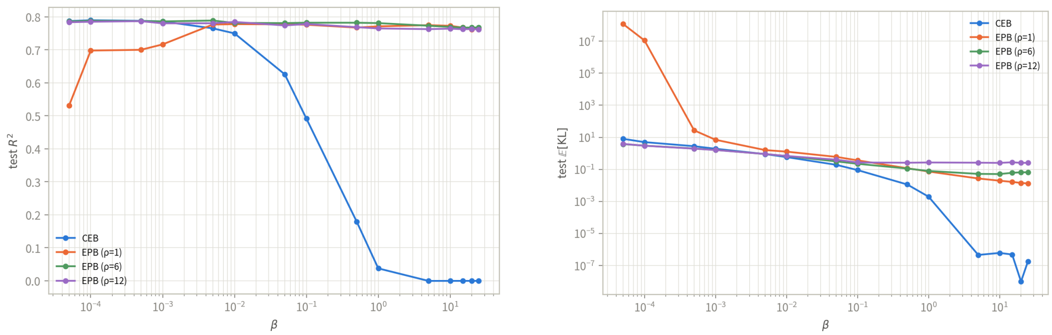

Figure 9: California Housing compression sweep (MLP). Left: test R^2; right: realized conditional KL over β. EPB maintains a task plateau through strong compression, while the $\rho = 1$ weak-compression points exhibit a large variance term.

Setting	CEB $I(X;Z)$	CEB Acc.	OPB config.	OPB $I(X;Z)$	OPB Acc.
MNIST, $\beta = 0.1$	2.53	98.2	$a = 1, \beta = 0.1$	2.55	98.1
MNIST, $\beta = 0.5$	1.96	86.0	$a = 1, \beta = 0.5$	2.01	97.8

Table 12: Matched-compression information-plane comparison (means; standard deviations are in the stored evaluation report).

B.5 Adversarial robustness details

Targeted PGD uses 20 steps, EOT-10 gradients, 10-sample MC predictions, and budgets $\varepsilon_\infty = 0.3$ and $\varepsilon_2 = 6$. Curves are shown only when clean stochastic accuracy exceeds 90%. At maximum compression, OPB $(a = 6, 12)$ reaches $31.1\%/30.5\%$ targeted success under L_∞, while transfer attacks from CEB to OPB reach at most 2.0%/3.4% under L_∞/L_2, far below OPB's white-box attack. The transfer result is a gradient-masking check rather than a causal geometry test.

C Related work

Information bottleneck and conditional compression. The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information–accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Comparisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

Information bottleneck for robust representations. Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates

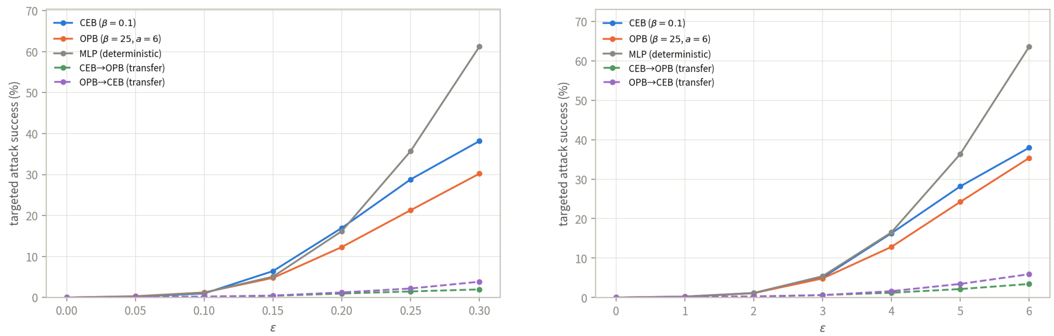

Figure 10: Transfer PGD curves on MNIST (gradient-masking check). Targeted adversaries are generated on one model and evaluated on another; CEB-to-OPB transfer remains far below OPB's own white-box attack. The targeted white-box curves are in the main text.

task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

Geometric classification representations. Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papyan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottleneck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

Table 13 summarizes this object-level distinction. The geometric primitive is shared with several preceding lines of work, but its role in the objective is different: OPB anchors are constrained class centers that serve simultaneously as Gaussian-prior means, KL targets, and readout anchors. In the shared-covariance canonical instance, the induced readout is an orthogonal classifier as a consequence of the Gaussian Bayes rule. It is not an ETF construction: OPB has Gram matrix $a^2 I_K$ (zero pairwise inner products), whereas a centered simplex ETF has constant negative off-diagonal inner products in a $(K-1)$-dimensional subspace. The frame is recomputed from the prior encoder and may rotate, rather than fixing a coordinate frame or optimizing a separate prototype table.

Distance-preserving regression and geometric priors. Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they generally constrain a deterministic feature map rather than the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The same axis is also the tied regression readout and the target of

Approach	Constrained object	Enters an IB KL?	Readout tied to geometry?	Stochastic posterior?
Prototype methods (Deng et al., 2014)	prototypes or features	No	Usually no/partial	Usually no
ETF / orthogonal classifiers (Xu et al., 2022; Ji et al., 2023)	classifier weights or feature directions	No	Yes	Usually no
Fixed or hierarchy-aware frames (Tong & Davis, 2023)	feature coordinates or a frame	No	Method-dependent	Usually no
CEB (Fischer, 2020)	label-conditional reference	Yes (conditional KL)	No (free head)	Yes
GPB (OPB / EPB)	label-conditional Gaussian prior means and covariance	Yes (conditional KL)	Yes (tied rule)	Yes

Table 13: Conceptual boundary between geometric representation methods and GPB. The entries describe the object constrained by the method, not whether a particular implementation may contain additional components. GPB reuses familiar geometric primitives while placing them in the conditional prior used by the bottleneck and using the same prior geometry for prediction.

the conditional KL. Thus EPB is a conditional-prior use of distance geometry, not a new distance-preserving primitive. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

Geometry-aware information bottlenecks. Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to- separation construction considered here.